



KrishiDrishti – AgriSmart AI

Model Training, Evaluation & Technical Report

Core feature: plant disease image classification

Model: EfficientNet-B0 **Classes:** 27 **Input:** 224×224 RGB

Final test accuracy: 99.86% **Final test Macro-F1:** 99.78%

Prepared from the supplied Google Colab notebooks and their recorded execution outputs. Report date: 15 September 2026.

1. Executive Summary

The supplied notebooks implement and evaluate a 27-class plant disease image classifier for KrishiDrishti. The final pipeline uses EfficientNet-B0, 224×224 image preprocessing, weighted cross-entropy, AdamW optimization, staged transfer learning, checkpointing, early stopping, and a packaged PyTorch model for inference.

The final fine-tuned model achieved **99.86% test accuracy**, **99.79% test macro precision**, **99.78% test macro recall**, and **99.78% test Macro-F1** on a held-out test set of 5,078 images. The best validation Macro-F1 reached **99.90%** at Stage 2 Epoch 4.

Dataset	Classes	Train	Validation	Test
PlantVillage	27	23,695	5,078	5,078
Test Accuracy	Macro Precision	Macro Recall	Macro-F1	Best Val Macro-F1
99.86%	99.79%	99.78%	99.78%	99.90%

Key engineering outcomes

- Dataset cleaning removed 21 invalid images and 21 exact duplicate files, leaving 54,284 valid images across 38 source classes before restricting to the 27 classes supported by the existing model.
- The 27-class working dataset contained 33,851 images and was split using a stratified 70/15/15 train/validation/test strategy.
- Stage 1 trained the classifier while freezing the EfficientNet feature extractor. Stage 2 unfroze the last three feature blocks and continued fine-tuning with a lower learning rate.
- The final model package stores the model state, class mapping, image size, validation score, and test metrics, supporting reproducible inference.

2. Dataset & Data Preparation

Source: PlantVillage, as identified in the notebook. The initial scan contained 54,305 images across 38 classes. After image validation and exact-duplicate removal, 54,284 images remained.

The existing KrishiDrishti checkpoint supports 27 classes. The notebook therefore filtered the cleaned dataset to those exact 27 classes, producing a final working dataset of **33,851 images**. Eleven source classes were excluded because they were not supported by the existing 27-class checkpoint.

Dataset split

A stratified split was used to preserve class proportions: approximately 70% training (23,695 images), 15% validation (5,078 images), and 15% test (5,078 images).

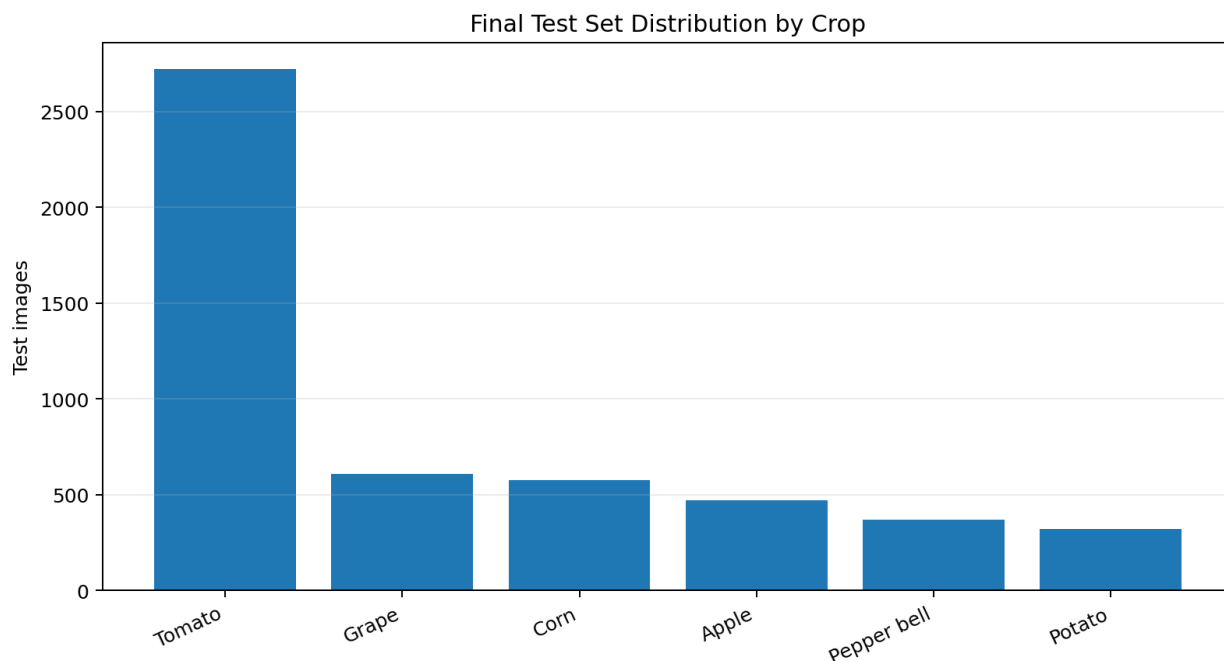


Figure 1. Test-set image distribution aggregated by crop from the recorded per-class test supports.

Image preprocessing and augmentation

All images are converted to RGB and resized to **224×224**. Training augmentation applies random horizontal flipping ($p=0.5$), random rotation ($\pm 15^\circ$), and ColorJitter (brightness, contrast, saturation 0.15; hue 0.03). Validation/test use resize, tensor conversion, and ImageNet normalization only.

Normalization uses mean [0.485, 0.456, 0.406] and standard deviation [0.229, 0.224, 0.225].

3. Model Architecture & Training Strategy

Backbone: EfficientNet-B0. The classifier head is replaced with a linear layer producing 27 class logits. The model is executed on a CUDA GPU during training; the notebook records a Tesla T4.

Two-stage transfer-learning strategy

Stage 1 — classifier adaptation: all EfficientNet feature layers were frozen; only the classifier remained trainable. AdamW was used with learning rate 0.001 and weight decay $1e-4$ for 5 epochs. The best recorded validation Macro-F1 was 0.9987.

Stage 2 — selective fine-tuning: the last three EfficientNet feature blocks were unfrozen while the classifier remained trainable. AdamW used learning rate $1e-5$ for the feature blocks and $1e-4$ for the classifier, with weight decay $1e-4$. Training was configured for up to 10 epochs with patience 3.

Loss and class balancing

The training objective is **weighted cross-entropy**. Class weights were calculated from the training-set class counts, giving additional emphasis to less represented classes. This is intended to reduce the impact of class imbalance.

Evaluation metrics

The notebook computes loss, accuracy, macro precision, macro recall, and macro F1. Macro averaging gives every class equal weight, which is useful when class frequencies differ.

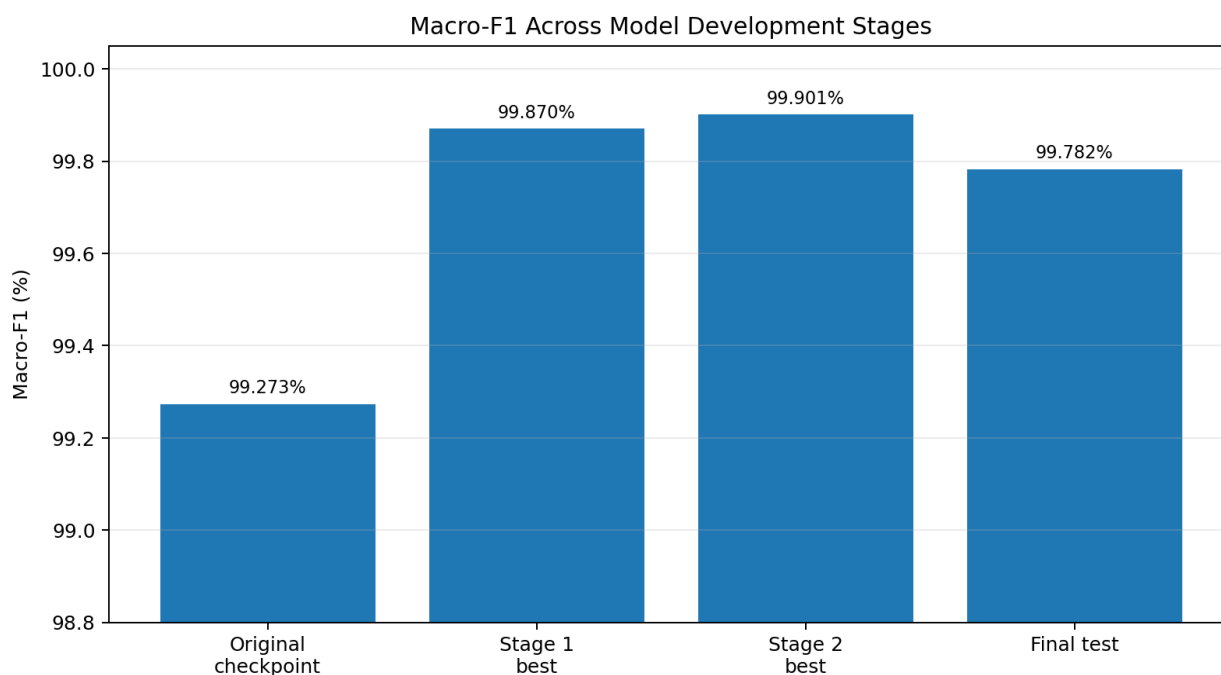


Figure 2. Macro-F1 across the original checkpoint, Stage 1, Stage 2, and final test evaluation. Values are taken from the notebook's recorded metrics.

4. Training Dynamics

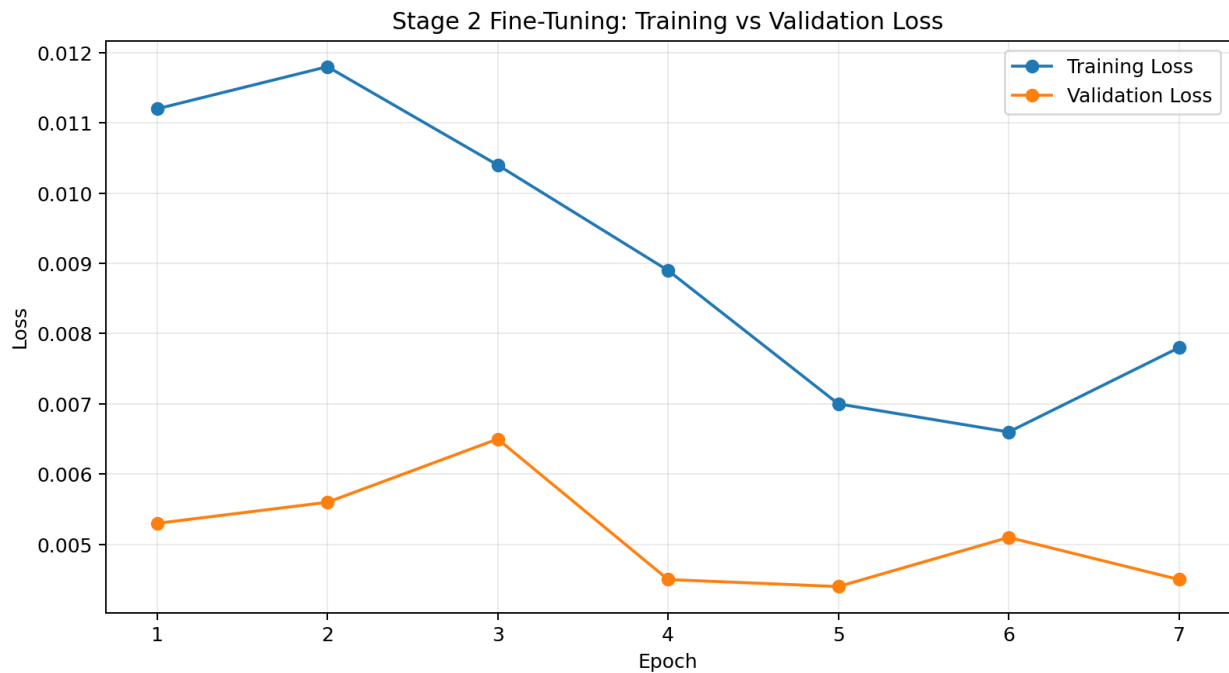


Figure 3. Stage 2 training and validation loss across the seven executed epochs.

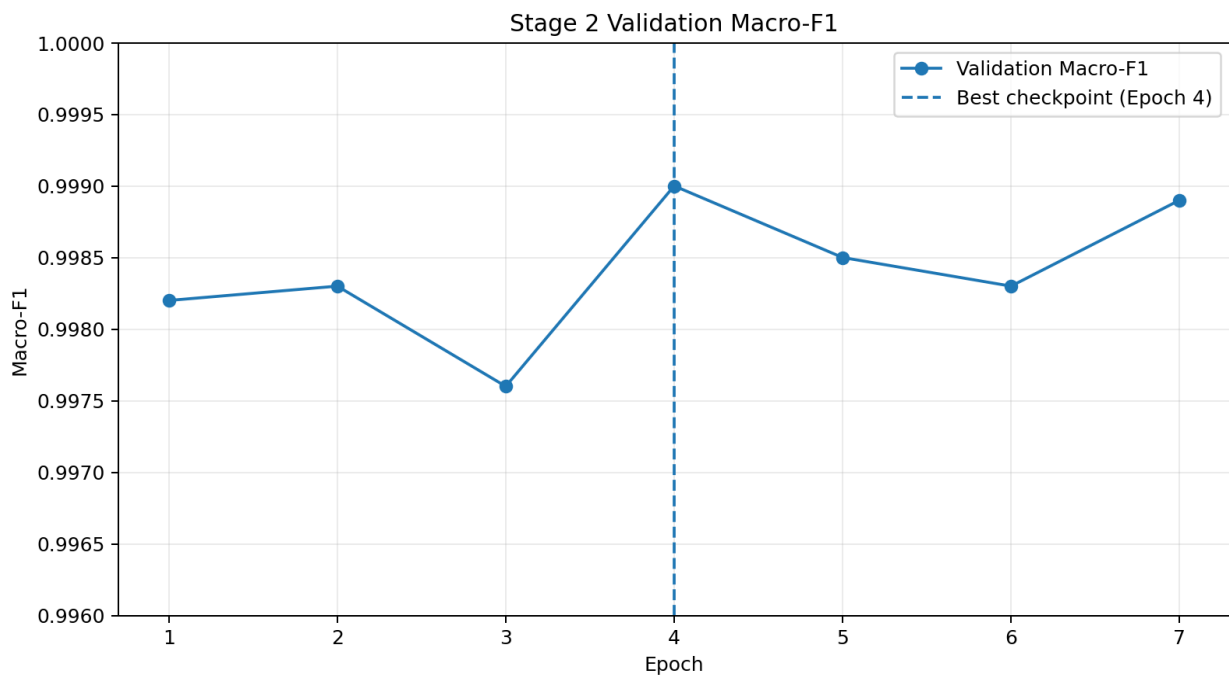


Figure 4. Stage 2 validation Macro-F1. The best checkpoint occurred at Epoch 4 (0.9990).

5. Training Accuracy & Validation Progression

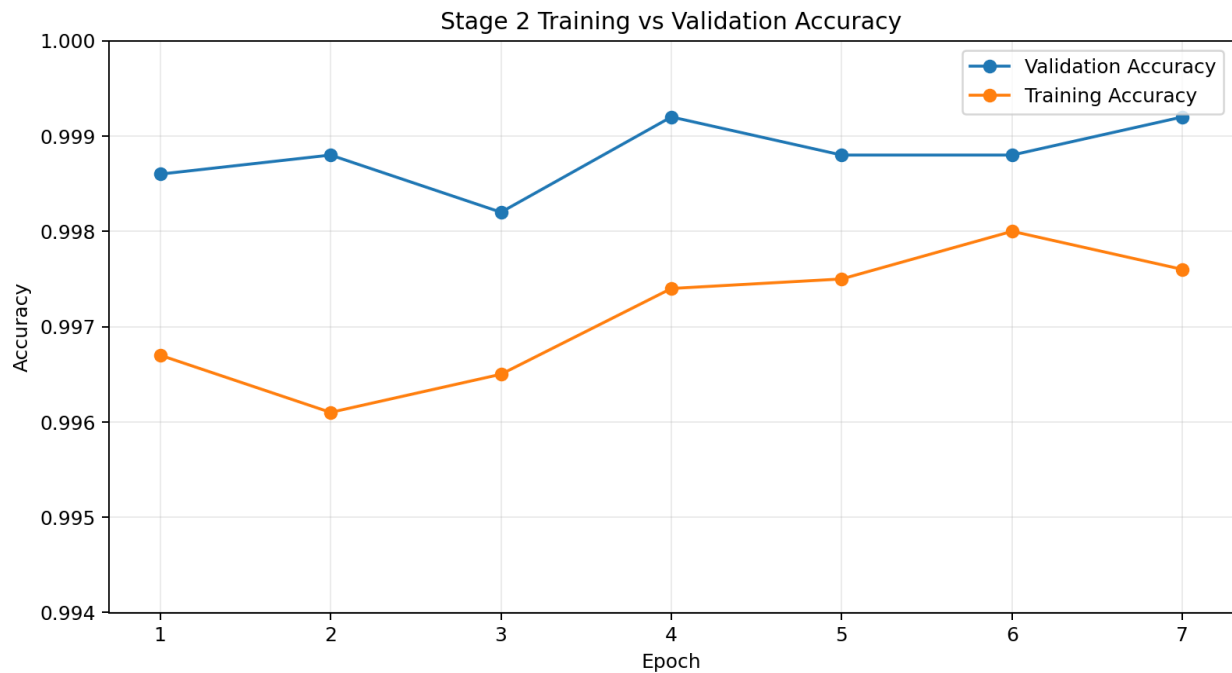


Figure 5. Stage 2 training and validation accuracy.

Stage 2 recorded seven executed epochs before early stopping. Validation Macro-F1 progressed from 0.9982 in Epoch 1 to 0.9990 in Epoch 4. Epochs 5–7 did not exceed the Epoch 4 best; the run stopped after the configured patience was exhausted.

Recorded Stage 2 history

Epoch	Train Loss	Val Loss	Train Acc.	Val Acc.	Val Macro-F1
1	0.0112	0.0053	0.9967	0.9986	0.9982
2	0.0118	0.0056	0.9961	0.9988	0.9983
3	0.0104	0.0065	0.9965	0.9982	0.9976
4	0.0089	0.0045	0.9974	0.9992	0.9990
5	0.0070	0.0044	0.9975	0.9988	0.9985
6	0.0066	0.0051	0.9980	0.9988	0.9983
7	0.0078	0.0045	0.9976	0.9992	0.9989

6. Final Test Evaluation

The best Stage 2 checkpoint was restored and evaluated once on the held-out test set of 5,078 images.

Metric	Value	Interpretation
Test loss	0.0101	Average weighted cross-entropy on the test set.
Test accuracy	99.86%	Fraction of test images assigned to the correct class.
Macro precision	99.79%	Mean precision across the 27 classes.
Macro recall	99.78%	Mean recall across the 27 classes.
Macro F1	99.78%	Harmonic mean of macro precision and macro recall.

Per-class performance

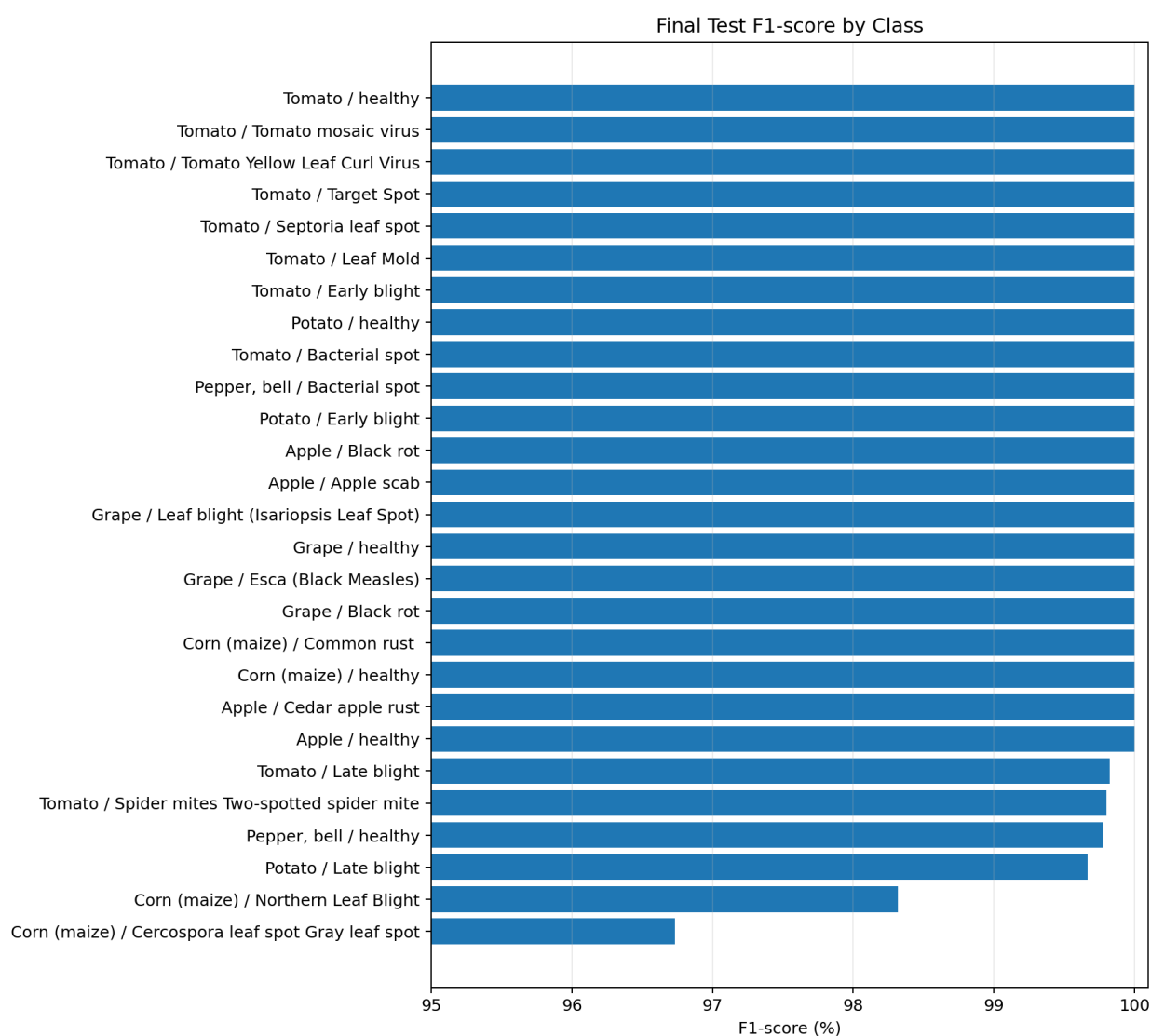


Figure 6. Final test F1-score by disease/healthy class. Most classes reached 100% F1; the lowest recorded class-level F1 was 96.73% for Corn Cercospora leaf spot / Gray leaf spot.

7. Detailed Per-Class Test Metrics

The following table reproduces the classification report recorded in the notebook for all 27 classes. Support is the number of held-out test images for each class.

Class	Precision	Recall	F1	Support
Apple / Apple scab	1.0000	1.0000	1.0000	94
Apple / Black rot	1.0000	1.0000	1.0000	93
Apple / Cedar apple rust	1.0000	1.0000	1.0000	41
Apple / healthy	1.0000	1.0000	1.0000	245
Corn (maize) / Cercospora leaf spot Gray leaf spot	0.9737	0.9610	0.9673	77
Corn (maize) / Common rust	1.0000	1.0000	1.0000	179
Corn (maize) / Northern Leaf Blight	0.9799	0.9865	0.9832	148
Corn (maize) / healthy	1.0000	1.0000	1.0000	174
Grape / Black rot	1.0000	1.0000	1.0000	177
Grape / Esca (Black Measles)	1.0000	1.0000	1.0000	208
Grape / Leaf blight (Isariopsis Leaf Spot)	1.0000	1.0000	1.0000	162
Grape / healthy	1.0000	1.0000	1.0000	63
Pepper, bell / Bacterial spot	1.0000	1.0000	1.0000	150
Pepper, bell / healthy	0.9955	1.0000	0.9978	222
Potato / Early blight	1.0000	1.0000	1.0000	150
Potato / Late blight	0.9934	1.0000	0.9967	150
Potato / healthy	1.0000	1.0000	1.0000	23
Tomato / Bacterial spot	1.0000	1.0000	1.0000	319
Tomato / Early blight	1.0000	1.0000	1.0000	150
Tomato / Late blight	1.0000	0.9965	0.9982	285
Tomato / Leaf Mold	1.0000	1.0000	1.0000	143
Tomato / Septoria leaf spot	1.0000	1.0000	1.0000	266
Tomato / Spider mites Two-spotted spider mite	1.0000	0.9960	0.9980	252
Tomato / Target Spot	1.0000	1.0000	1.0000	210
Tomato / Tomato Yellow Leaf Curl Virus	1.0000	1.0000	1.0000	803
Tomato / Tomato mosaic virus	1.0000	1.0000	1.0000	56
Tomato / healthy	1.0000	1.0000	1.0000	238

8. Confusion Matrix

The notebook generated a 27×27 confusion matrix on the final test predictions. The matrix provides a class-by-class view of correct classifications and residual confusions.

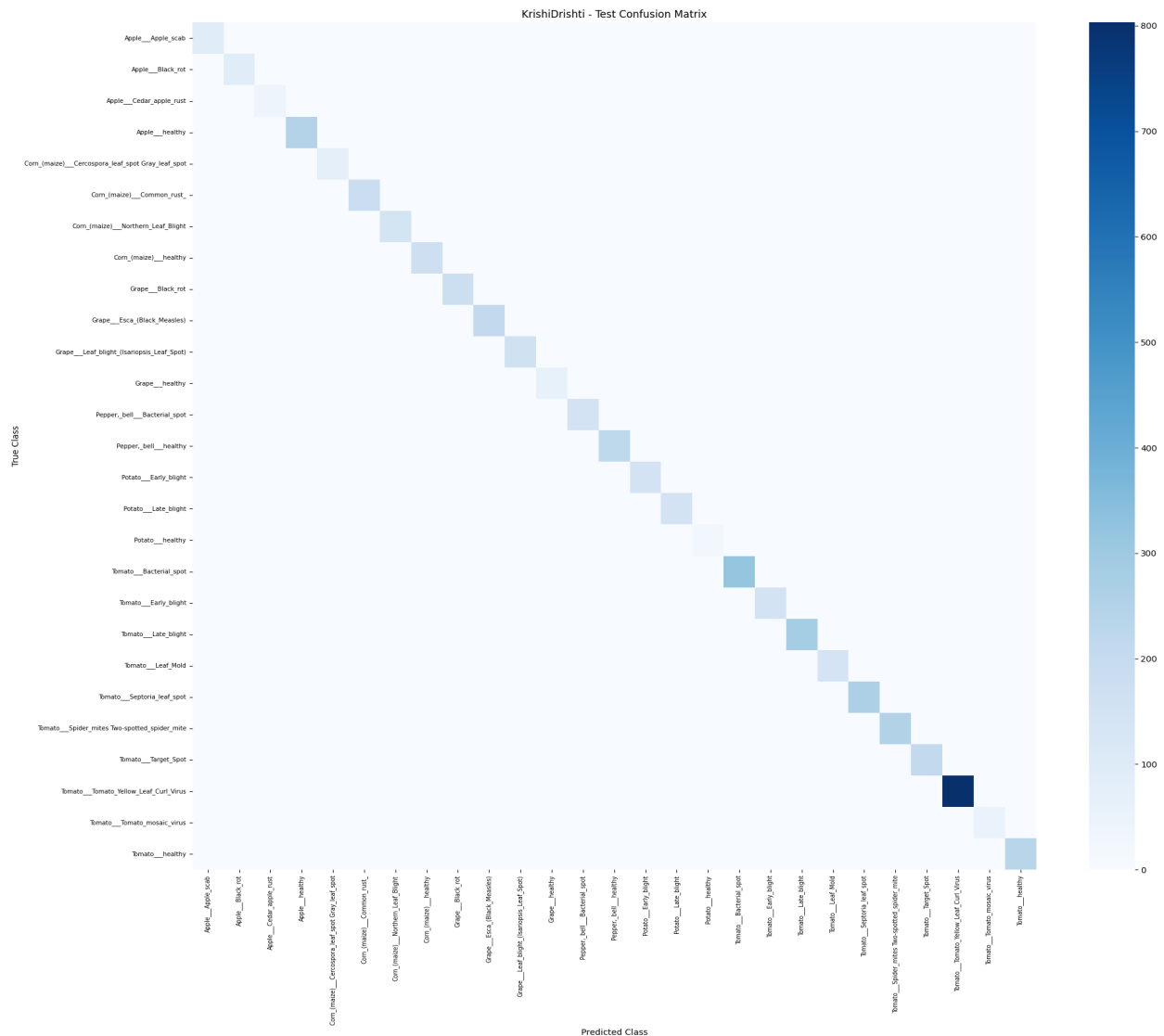


Figure 7. Confusion matrix generated directly by the supplied notebook for the final test evaluation.

Because the matrix is 27×27, small off-diagonal counts are visually dense. The class-level precision/recall/F1 table on the previous page should be used alongside the matrix when interpreting individual classes.

9. Inference & Model Packaging

The notebooks implement a reusable inference pipeline. A saved PyTorch package contains the EfficientNet-B0 state dictionary, 27 class names, class-to-index and index-to-class mappings, image size, best validation Macro-F1, and final test metrics.

The inference transform matches validation/test preprocessing: resize to 224×224, convert to tensor, and apply ImageNet normalization.

The prediction interface returns the predicted class and confidence. A top-3 prediction function is also implemented. A farmer-friendly mapping converts raw class labels into crop, disease/healthy condition, status, and confidence.

Recorded sample inference

The notebook records an example prediction of **Corn — Northern Leaf Blight** with **97.04%** confidence. A separate direct prediction example recorded Corn Cercospora leaf spot / Gray leaf spot at 96.90% confidence.

Final model artifact

The final model was saved as **krishidrishti_efficientnet_b0_final.pth**. The notebook also verifies the saved artifact and reports 27 classes, 224-pixel input size, 99.90% best validation Macro-F1, 99.78% test Macro-F1, and 99.86% test accuracy.

10. Robustness Testing & Limitations

The supplied notebook includes an **invalid-image robustness test** that is designed to upload non-leaf images (for example, people, animals, cars, buildings, food, and landscapes), report the top-3 predictions, and flag predictions above 90% confidence for investigation.

Important: the notebook's recorded output does not contain completed results from that robustness test. Therefore, this report does not claim an invalid-image rejection rate or any robustness score.

Evaluation scope and limitations

1. The reported metrics are based on a held-out PlantVillage test split and should not be interpreted as field-level accuracy on arbitrary smartphone photographs.
2. The 27-class scope excludes 11 classes present in the original 38-class dataset scan because the existing model did not support them.
3. The final confidence value is the model's softmax probability; it is not, by itself, a calibrated probability of real-world correctness.
4. The notebook evaluates image classification, not treatment prescription, yield prediction, weather forecasting, or agronomic diagnosis beyond the mapped disease information.

Recommended next validation steps

For a production-grade release, add a geographically and visually diverse field dataset, test across different phones/lighting/backgrounds, quantify out-of-distribution rejection, calibrate confidence thresholds, and report performance on genuinely unseen farms and acquisition conditions.

11. Reproducibility Checklist

The supplied notebooks record the following configuration needed to reproduce the reported training/evaluation design:

- Framework: PyTorch / torchvision
- Backbone: EfficientNet-B0
- Input: 224×224 RGB
- Dataset: PlantVillage-derived working set
- Working classes: 27
- Split: stratified 70/15/15
- Batch size: 32
- Stage 1: classifier-only, AdamW, LR 1e-3, 5 epochs
- Stage 2: last 3 feature blocks + classifier, AdamW, LR 1e-5 / 1e-4, weight decay 1e-4, patience 3
- Loss: weighted CrossEntropyLoss
- Normalization: ImageNet mean/std
- Selection criterion: validation Macro-F1
- Final evaluation: 5,078-image held-out test set

Source basis

This report was generated solely from the supplied **stage1-2(1).ipynb** and **core(1).ipynb** notebooks, including their recorded execution outputs, plus the supplied KrishiDrishti logo image. Where an experiment was implemented but its execution output was absent, the report explicitly marks the result as unavailable rather than estimating it.

End of report